

WALMART FRONTLINE AI TRANSFORMATION

Individual Contribution Statement

ALY-6130 | Group 9 | Northeastern University | Vancouver Campus | June 25th 2026

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Primary Contributions

The following table documents my specific and primary contributions to the Walmart Frontline AI Transformation ERM project across all modules of ALY 6130. Each contribution reflects my individual area of responsibility within the team.

Module	Deliverable	Specific Contribution
M3	Risk Calculation Sheet	Led the development of the risk calculation sheet (Conservative sheet), defining the probability of occurrence and impact scales, scoring all 30 risks using the Likelihood × Impact methodology across six risk categories (Financial, Operational, Competitive, Reputational, Technology, Compliance), and establishing the High/Medium/Low threshold definitions (Low: 1-14, Medium: 15-39, High: 40-81).
M3	Excel Heatmap	Developed the Excel-based risk heatmap (HeatMap Excel sheet) with correct X-axis (Likelihood Score) and Y-axis (Impact Score) mapping using the exact scale values from the Conservative sheet (Likelihood: 1,3,5,7,9; Impact: 1,2,4,6,8,9). Implemented color-coded severity classifications (Red=High, Yellow=Medium, Green=Low) with all risk IDs plotted correctly across the matrix cells.
M4	Python Notebook Development	Developed the complete Python Jupyter notebook (walmart_fixed.ipynb) incorporating: real Walmart financial data extraction via the SEC EDGAR API; market share trend analysis for R1; customer attrition analysis for R2; inventory turnover improvement analysis for R3; Expected Loss bar chart visualizations; and the I&W warning probability dashboard. Uploaded and shared the notebook with the team for review and modification.
M4	Monte Carlo Simulation	Implemented the Monte Carlo simulation component of the Python notebook, running 10,000 iterations per risk using Beta distribution sampling for probability uncertainty and Normal distribution sampling for impact uncertainty (±20% ranges). Generated P10/Mean/P90 distribution outputs and right-skewed loss distribution histograms for R5 and R6, with symmetric gain distribution for R3.
M4	Random Forest ML Model	Built and validated the Random Forest classification model within the Python notebook, trained on a 500-record synthetic retail AI risk dataset. Achieved 92% classification accuracy on the 25% held-out test set. Developed the confusion matrix visualization, feature

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M6	Presentation 23 Slides	importance bar chart, and Walmart specific risk warning probability outputs (R1: 69.2%, R2: 21.6%, R3: 3.9%, R5: 85.3%, R6: 78.4%). Led the development of the Group 9 presentation (Walmart Frontline AI Transformation.pptx) covering all 23 slides: Introduction and agenda, Business context and integrated situation assessment, ERM framework workflow, Risk identification and heatmap visualization, KRI early warning system, Quantitative assessment and I&W analysis, Monte Carlo financial exposure modeling, ML validation results, Risk response strategy, Residual risk analysis, Strategic recommendations, and Conclusion.

Shared Team Contributions

In addition to the primary contributions documented above, I participated equally with all team members in the following collaborative activities:

- Weekly team discussions and structured brainstorming sessions that informed risk identification, scoring, and response strategy decisions across all modules.
- Peer review of all deliverables before submission, providing feedback on content accuracy, APA citation format, and professional language standards.
- GitHub repository management, reviewing and approving pull requests, contributing to the collaborative version control workflow throughout the course.
- Joint review and editing of the final Signature Assessment report to ensure consistency, accuracy, and alignment across all sections.